

Stack: | MinStack:



[155. Min Stack](#)

Medium



Topics



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Hint

Design a stack that supports push, pop, top, and retrieving the minimum element in constant time.

Implement the `MinStack` class:

- `MinStack()` initializes the stack object.
- `void push(int value)` pushes the element `value` onto the stack.
- `void pop()` removes the element on the top of the stack.
- `int top()` gets the top element of the stack.
- `int getMin()` retrieves the minimum element in the stack.

You must implement a solution with $O(1)$ time complexity for each function.

Example 1:

Input

```
["MinStack", "push", "push", "push", "getMin", "pop", "top", "getMin"]  
[[], [-2], [0], [-3], [], [], [], []]
```

Output

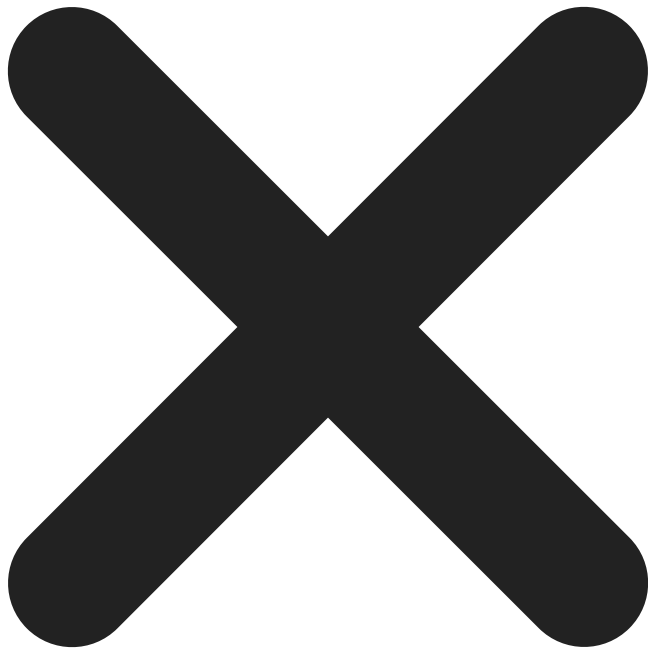
```
[null, null, null, null, -3, null, 0, -2]
```

Explanation

```
MinStack minStack = new MinStack();  
minStack.push(-2);  
minStack.push(0);  
minStack.push(-3);  
minStack.getMin(); // return -3  
minStack.pop();  
minStack.top();    // return 0  
minStack.getMin(); // return -2
```

Constraints:

- $-2^{31} \leq \text{val} \leq 2^{31} - 1$
- Methods `pop`, `top` and `getMin` operations will always be called on **non-empty** stacks.
- At most $3 * 10^4$ calls will be made to `push`, `pop`, `top`, and `getMin`.



Seen this question in a real interview before?

1/6

Yes

No

Accepted

2,758,989/4.7M

Acceptance Rate

58.2%

 Topics



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Hint 1



Consider each node in the stack having a minimum value. (Credits to @aakarshmadhavan)



Discussion (315)



</>



Discussion Rules

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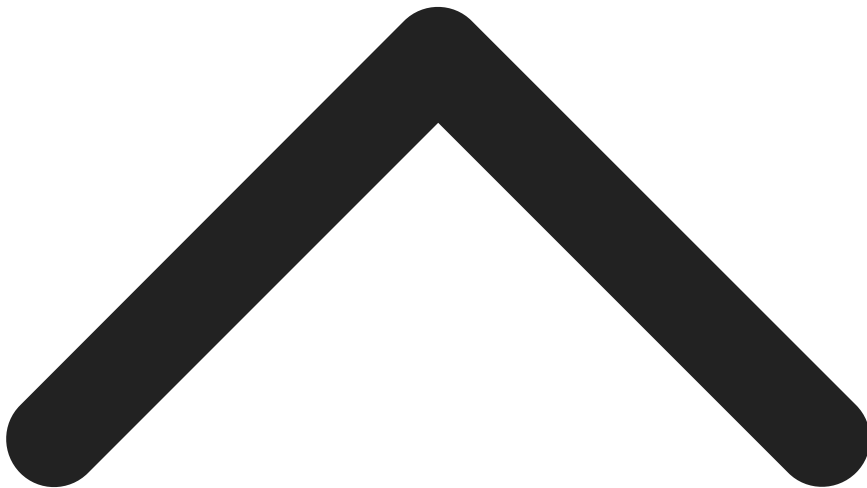
1. Please don't post **any solutions** in this discussion.
2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.

3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.



[Hristo Yankov](#)

Dec 01, 2019



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1.3K



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[Sayan B](#)



Jul 27, 2023

With this problem, today I mark the day of solving 100 mediums!



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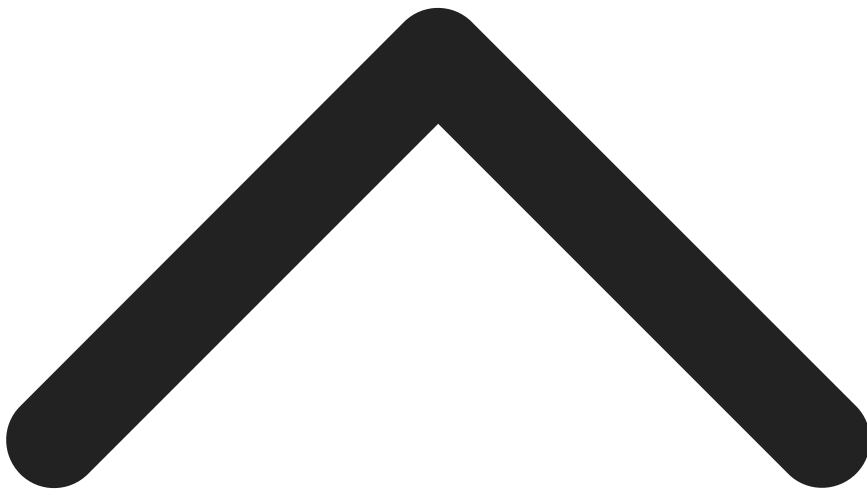


[bikcrum](#)



Nov 19, 2022

I would have published a paper if it was possible to retrieve minimum element at $O(1)$. As far as I know the least is $\log(n)$ using heap.



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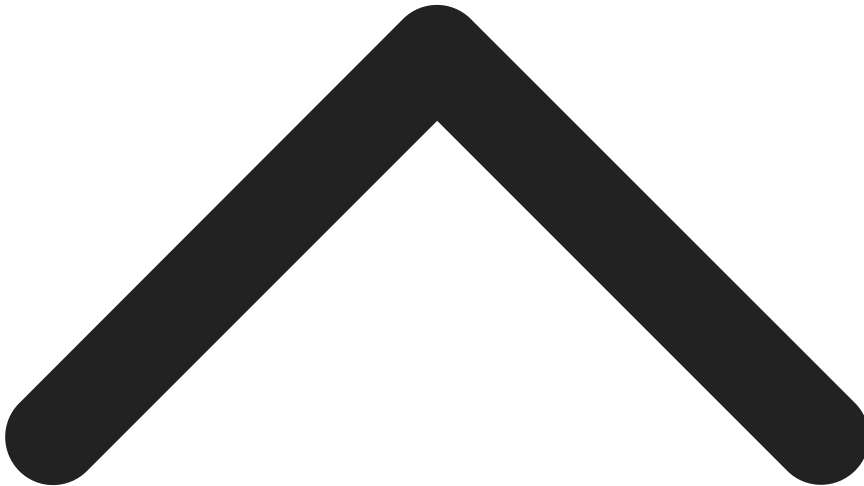
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[lussierjames13](#)

Dec 22, 2022

Try associating each value in the stack with a minimum value



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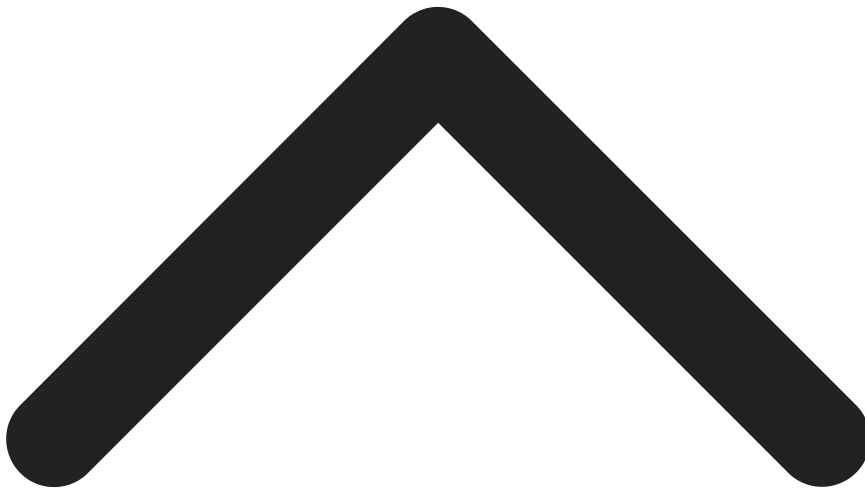


[Guilherme Zardo](#)



May 17, 2023

Am I the only one who thinks the problem should be more clear? They don't say that we do not need to remove the minimum element from the stack, only "retrieve it". But how am I supposed to know what "retrieving an element" is? It doesn't help that they used "gets", "retrieves" and "removes" in the explanation for three situations. I spent a lot of time trying to figure out a way to remove the minimum element from the stack in constant time.



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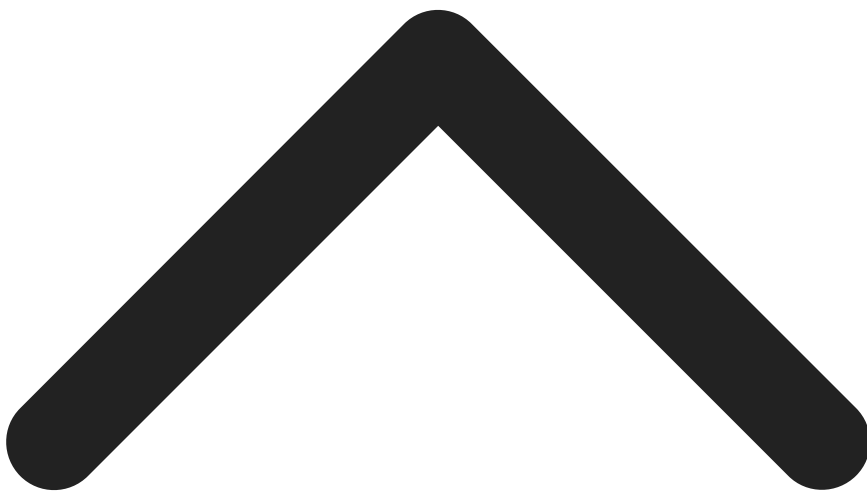
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[Amd](#)

Apr 08, 2023

Notice that getMin does not remove it from stack.



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Tip



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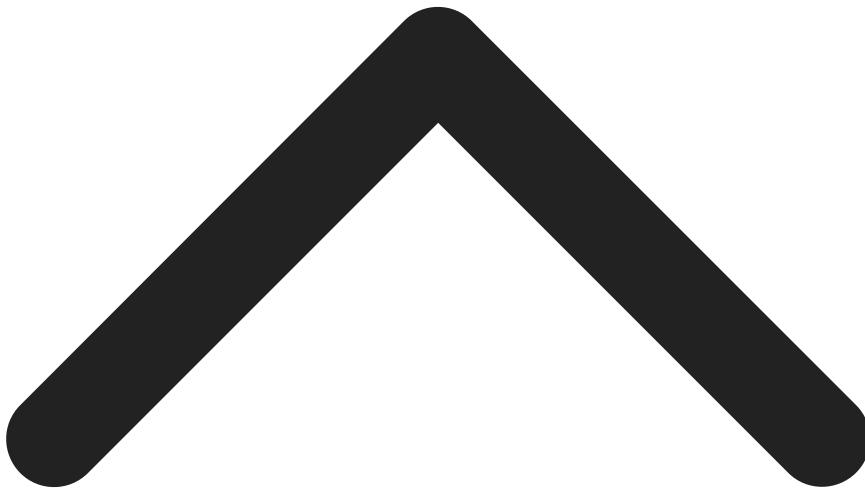
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[Bill Gewrgoulas](#)

Mar 25, 2023

Super happy i could this on my own, excellent problem!



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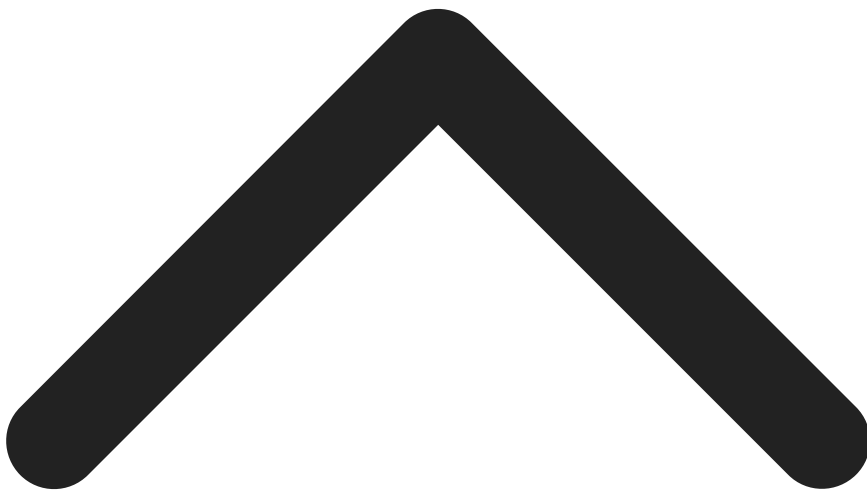
[Felix Fang](#)



Jan 23, 2024

What is the point of designing a stack using stack?

It's like solving a sorting problem using "sort()"



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[Akshaj Vadakkath Joshy](#)



Jun 20, 2023

4	-9
0	-9
-9	-9
2	-5
-5	-5
10	-5
-2	-3
8	-3
-3	-3
5	5

Try implementing using two stacks. One stack keeps track of elements in order, while the other stack keeps track of the minimum elements.

LOOK DOWN TO UP ITS A STACK

^
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Tip

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⬇
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[Murph](#)



Feb 09, 2024

The trick here is to understand that, since it is a stack and we don't care about indexes, we don't need to worry about sorting...

Example, let's say we push the elements -2, 0, -1, in that order.

As we see, the minimum value is -2, so, the trick is to notice that we can't ever get to the element with value -2 unless we pop the stack two times. I was giving it too much thought and wasted an hour on this, as my thought of process was: "If we pop -2, how are we going to know that -1 is the next minimum value?" Until I realized we don't need to do that, as -2 is the first value we inserted, so in order for -1 to be the next minimum value is to pop the stack until -2 stops existing, in which case -1 would have already stopped existing as well.



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